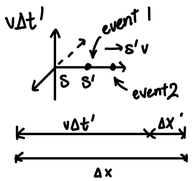
Newtonian

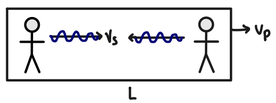
↳ Galilean Transformations: $\Delta x = \Delta x' + v\Delta t'$

$\Delta t = \Delta t'$

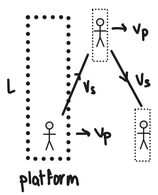


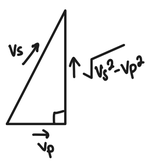
↳ $a_1 \equiv \frac{dv_1}{dt_1} = \frac{d(v_2+v_{rel})}{dt_1} = \frac{dv_2}{dt_2} + 0 = a_2$

parallel motion:



$t_1 = \frac{L}{v_s - v_p} + \frac{L}{v_s + v_p} = \frac{2L\,v_s}{v_s^2 + v_p^2}$





$t_2 = \frac{2L}{\sqrt{v_s^2 - v_p^2}}$

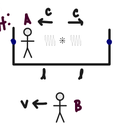
Intro → Relativity

postulates:

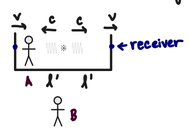
1) All inertial reference frames are 'equivalent' *(S = S')

2) The speed of light is the same from any inertial reference frame – c

A's event:



B's event:



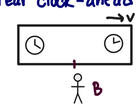
time to hit receiver from B's pov

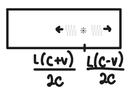
↳ $t_L = \frac{l'}{c+v}$, $t_R = \frac{l'}{c-v}$: $t_L < t_R$

⇒ not simultaneous!

$* \Delta t = \gamma \Delta t_0$

"rear clock-ahead" effect:





extra distance traveled by $\mathcal{O}_L$: $\frac{L(c+v)}{2c} - \frac{L(c-v)}{2c} = \frac{Lc + Lv - [Lc - Lv]}{2c}$

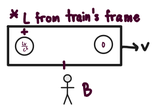
↳ $\frac{2Lv}{2c} = \frac{Lv}{c}$

time $= \frac{\Delta x}{v}$

so, speed ↳ $\frac{Lv}{c^2}$

$t_R - t_L$: $\frac{l'}{c-v} - \frac{l'}{c+v} \Rightarrow \frac{l'}{c-v}\frac{(c+v)}{(c+v)} - \frac{l'}{c+v}\frac{(c-v)}{(c-v)}$

↳ $\frac{l'(c+v)}{c-v(c+v)} - \frac{l'(c-v)}{c+v(c-v)} \Rightarrow \frac{\cancel{cl'} + vl'}{c^2 + \cancel{vc} - \cancel{vc} - v^2} - \frac{\cancel{cl'} - vl'}{c^2 - \cancel{vc} + \cancel{vc} - v^2}$



$\Rightarrow \frac{2vl'}{c^2 - v^2} \Rightarrow \frac{v(2l')}{c^2 - v^2}$: $2l' = L$: $\frac{Lv}{c^2 - v^2} \simeq \frac{Lv}{c^2}$

(assuming $v \lll c$)

object velocity: 2 reference frames:

S – reference object's "rest frame"

S' – moving object's rest frame

} both S or S' can be moving; choice of S will det. relative frame

3 reference frames w/ Δv:

$$u = \frac{v + u'}{1 + \frac{vu'}{c^2}}$$

speed of moving frame S'

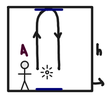
speed of object measured in moving frame S'

speed of object measured in rest frame S

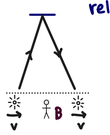
time dilation:

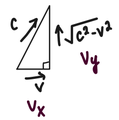
round trip time observed by A

↳ $t_A = \frac{2h}{c}$

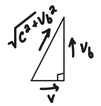


relativistic:





non-relativistic:



so, from B's pov – the round trip time observed

↳ $\frac{2h}{\sqrt{c^2 + v^2}}$

→ putting this together we get the following

⇒ $\frac{t_B}{t_A} = \frac{2h}{\sqrt{c^2 + v^2}} \Big/ \frac{2h}{c} \Rightarrow \frac{c}{\sqrt{c^2 + v^2}} = \frac{1}{\sqrt{1 - (v^2/c^2)}}$

which we define by $\gamma = \frac{1}{\sqrt{1 - (v^2/c^2)}}$

and arrive at the equivalence: $t_B = \gamma t_A$ where, for example γ @ $c/2$, $\gamma = 2/\sqrt{3}$